

Porous PI/h-BN films made by NIPS over the full 0–70 wt% h-BN range 3-phase Lewis-Nielsen model unifies dielectric and thermal behavior Pore shape factor $A = 0.08$ fingerprints NIPS sponge vs. spherical pores Single-parameter thermal fit: $\lambda_{\text{BN,eff}} = 6.8 \text{ W m}^{-1} \text{ K}^{-1}$, interface-limited 70 wt% h-BN locus is the global Pareto frontier of the system

Multifunctional Porous Polyimide/hexagonal boron nitride Composite Films Fabricated by Non-Solvent Induced Phase Separation: Coupled Experimental Characterization and 3-Phase Lewis–Nielsen Composite Modeling description of dielectric and thermal properties

[Author names]^{a,*}

^aDepartment of Chemical and Biomolecular Engineering, Yonsei University, Seoul, Republic of Korea

Abstract

Polymer films for next-generation electronic packaging must combine a low dielectric constant with sufficient thermal conductivity and dimensional stability. Most high-thermal-conductivity fillers are electrically conductive and destroy the dielectric character of the film, so ceramic fillers such as hexagonal boron nitride (h-BN) are preferred, although their affinity with polymer matrices is often limited. Porous polymers offer excellent low-permittivity dielectrics, but porosity suppresses thermal conductivity and mechanical strength, and high filler loadings are difficult to achieve in porous architectures. Here, porous polyimide (PI)/h-BN composite films spanning 0–70 wt% h-BN were fabricated by non-solvent induced phase separation (NIPS), and hot-pressing was applied to restructure the pore network, improving mechanical strength and raising thermal conductivity while retaining the low permittivity of the porous state. The dielectric, thermal, thermomechanical, and mechanical properties of all ten films were measured, and the full dataset was then interpreted and verified within a single calibrated three-phase Lewis–Nielsen framework treating each film as a PI, h-BN, and air system. The fitted pore shape factor ($A_{\text{pore}} = 0.08$) fingerprints the bicontinuous NIPS sponge structure, and the framework identifies 70 wt% h-BN with ~~controlled 50–60%~~ porosity as the ~~optimal design point, where the film achieves most favorable design region: the non-pressed film reaches~~ $\epsilon_r = 1.28$ ~~and~~ $\lambda = 0.31 \text{ W m}^{-1} \text{ K}^{-1}$, and ~~CTE = 10.1 ppm/K. the hot-pressed film combines~~ $\epsilon_r = 1.62$ ~~and~~ $\lambda = 0.28 \text{ W m}^{-1} \text{ K}^{-1}$ ~~with a CTE of 10.1 ppm K⁻¹. These results establish the pore shape factor as a microstructural design parameter for porous ceramic-filled polymer films and provide a calibrated basis for selecting composition and porosity in NIPS-derived packaging films.~~

Keywords: Polymer-matrix composites, Hexagonal boron nitride, Porous film, Lewis–Nielsen model, Dielectric properties, Thermal conductivity

1. Introduction

The convergence of high-density microelectronics, millimeter-wave fifth-generation (5G) communication, and electric-vehicle power electronics is ~~driving a fundamental shift in~~ ~~changing~~ the property requirements for polymer substrate and packaging materials. Contemporary device architectures demand films that simultaneously satisfy three stringent criteria: low dielectric constant ($\epsilon_r < 2.5$) to minimize signal delay and cross-talk at GHz–THz operating frequencies; adequate thermal conductivity ($\lambda > 0.2 \text{ W m}^{-1} \text{ K}^{-1}$) to dissipate heat fluxes that increasingly exceed 100 W cm^{-2} in advanced packages; and dimensional stability,

*Corresponding author.

Email address: [corresponding e-mail] ([Author names])

expressed as a coefficient of thermal expansion (CTE $\sim 20 \text{ ppm/K}$) matched to copper metalization ($\sim 17 \text{ ppm/K}$) and silicon substrates ($\sim 3 \text{ ppm/K}$) to prevent fatigue delamination under thermal cycling [1, 2, 3]. No single existing polymeric material satisfies all three criteria without modification, and the strategies used to improve one property typically degrade another—a fundamental challenge in multifunctional material design.

Aromatic polyimide (PI), synthesized from dianhydride and diamine monomers, has served as the foundational dielectric substrate material in the electronics industry for decades owing to its combination of high thermal stability (decomposition temperature $> 500^\circ\text{C}$), outstanding mechanical properties, chemical resistance, and electrical insulation [4]. The canonical PMDA–ODA system employed in the present work exhibits a dielectric constant of $\sim 3.4\text{--}3.8$, thermal conductivity of $\sim 0.12 \text{ W m}^{-1} \text{ K}^{-1}$, and CTE of $\sim 40\text{--}60 \text{ ppm/K}$ for dense films—properties that increasingly, values that fall short of 5G and power-electronics requirements [5]. Multiple molecular design strategies have been explored to intrinsically reduce the dielectric constant of PI, including incorporation of fluorinated monomers (hexafluoropropylidene groups), introduction of bulky pendant groups that increase free volume, and use of low-polarizability non-conjugated diamine structures [6, 7]. While these approaches can achieve $\epsilon_r < 3.0$, they typically require specialized monomers and complex synthesis, and they do not address the thermal conductivity deficit.

Porous structure engineering through non-solvent induced phase separation (NIPS) offers a distinct and complementary approach to dielectric reduction. In NIPS, a poly(amic acid)/solvent (NMP) solution is cast onto a substrate and immersed in a non-solvent bath, typically a lower-surface-energy solvent such as acetone, water, or alcohols. Rapid exchange of solvent and non-solvent across the film–bath interface drives thermodynamic instability—(spinodal decomposition or nucleation-and-growth—nucleation and growth), yielding a bicontinuous or sponge-like porous architecture that is preserved upon thermal imidization [8, 9, 10]. Because air ($\epsilon_{\text{air}} = 1.0$) has the lowest dielectric constant of any material, incorporating large air volume fractions through NIPS can dramatically reduce the effective film permittivity far below any molecular modification strategy reduces the effective permittivity well below what molecular modification alone achieves. Reported values span $\epsilon_r = 1.51\text{--}2.48$ for NIPS-derived porous PI and related engineering-polymer films with porosities up to 75%, with the dielectric constant continuously tunable by adjusting polymer concentration, non-solvent identity, and immersion conditions [7, 11, 12]. The approach is also directly scalable and compatible with roll-to-roll processing, unlike aerogel or freeze-drying methods that require supercritical drying or cryogenic equipment [13].

The thermal and dimensional limitations of porous PI can be addressed by incorporation of hexagonal boron nitride (h-BN) as a multifunctional inorganic filler. h-BN—also known as ‘white graphite’ due to its structural analogy with graphite—is a wide-bandgap ceramic ($E_g \approx 6 \text{ eV}$) with a layered hexagonal crystal structure that confers exceptional $E_g \approx 6 \text{ eV}$ whose layered hexagonal structure confers strongly anisotropic thermal conductivity: up to $400 \text{ W m}^{-1} \text{ K}^{-1}$ in-plane and $\sim 50 \text{ W m}^{-1} \text{ K}^{-1}$ through-plane, placing it among

the most thermally conductive electrically insulating materials known [14]. The dielectric constant of h-BN (~ 6.0 ; $\varepsilon_{\perp} = 7.04$, $\varepsilon_{\parallel} = 4.95$) is substantially lower than competing thermally conductive ceramics such as aluminum nitride ($\varepsilon_r \approx 9$) and silicon carbide ($\varepsilon_r \approx 10$), making it the filler of choice when dielectric performance must be preserved alongside thermal enhancement [15]. Furthermore, the in-plane CTE of h-BN ($\sim 1.5 \text{ ppm/K}$) and its high in-plane elastic modulus ($\sim 200\text{--}700 \text{ GPa}$) make it an effective dimensional stabilizer in polymer composites through elastic constraint mechanisms [16, 17].

Prior studies of dense (non-porous) PI/h-BN composite films — where the PI matrix is not porous — have demonstrated thermal conductivity values of $0.5\text{--}6.2 \text{ W/mK}$ at BN $\sim 6.2 \text{ W m}^{-1} \text{ K}^{-1}$ at h-BN loadings of 10–60 wt%, typically at the cost of increased dielectric constant ($\varepsilon_r = 3.5\text{--}5.0$) [18, 19, 20]. Yang et al. [21] fabricated porous BN-h-BN/PI composite films by combining high-internal-phase Pickering emulsion (HIPPE) and hot-pressing, achieving simultaneously low ε_r and high thermal diffusivity. However, the spherical emulsion-templated pores in HIPPE films are geometrically distinct from the bicontinuous sponge pores generated by NIPS, resulting in fundamentally different pore structure contributions so the two pore architectures contribute differently to composite properties — a distinction that. This distinction has not previously been quantified within a unified-single modeling framework. Porous PI films have likewise shown enhanced dielectric stability at elevated temperature [22], while electrospinning-hot-press routes have produced dense BN-h-BN/PI composites with outstanding-high thermal stability [15]. Missing from the literature is a quantitative model that relates dielectric constant and thermal conductivity simultaneously to both BN-h-BN loading and porosity, provides physically interpretable composite microstructure parameters, and generates actionable design guidance for property optimization in porous PI/BN systems specifically h-BN systems fabricated by NIPS.

The Lewis–Nielsen model, originally formulated by Lewis and Nielsen [23, 24] as (LN) model [23, 24], a modification of the Halpin–Tsai equations, provides exactly this capability. It predicts the effective scalar transport property of a two-phase particulate composite as a function of filler volume fraction, filler–matrix property contrast, filler shape (through an Einstein coefficient A), and maximum packing fraction $\phi_{\max}$. By applying the model sequentially to a three-component system — first mixing, first mixing the PI matrix with h-BN filler to obtain a dense composite, and then introducing air pores — a 3-phase, a three-phase LN model is constructed that uses measured porosities and BN-h-BN loadings as inputs and yields fitted parameters that directly characterize the h-BN nanoplatelet geometry and the nature of the NIPS-generated pore architecture [25]. Analogously, the Turner model [26] handles CTE through elastic-modulus-weighted mixing, and the Gibson–Ashby model [27] provides the standard porosity-scaling law for mechanical properties of open-cell porous solids.

In this work, we present the first-a systematic experimental and modeling study of NIPS-derived porous PI/h-BN composite films across the full composition range (0–70 wt% h-BN) in both non-pressed (NP) and hot-pressed (HP) states. The central contribution is the calibrated 3-phase Lewis–Nielsen framework that integrates all measured properties — dielectric constant, thermal conductivity, CTE, and tensile

strength—into a unified quantitative description a calibrated three-phase LN framework that describes the measured dielectric constant and thermal conductivity with one shared microstructural parameterization, extracts physically meaningful composite microstructure interpretable composite parameters, and generates 2D design maps, an Ashby-style property diagram, and a global Pareto-frontier analysis to guide experimental optimization. Critically, the design maps for simultaneous property optimization. The fitted pore shape factor $A_{\text{pore}} = 0.08$ provides a quantitative microstructure fingerprint that distinguishes NIPS sponge pore geometry pores from the spherical pores of HIPPE-based composites, the HIPPE composites; a deliberately parsimonious single-parameter thermal calibration avoids the over-fitting pitfalls common to multi-parameter composite models, and the model-guided analysis directly; and the calibrated model prescribes the porosity required to satisfy demanding meet dual-property specifications. We evaluate the framework against ten films to show that porosity and pore geometry, not filler geometry, govern the dielectric response, whereas thermal conductivity is gated by filler loading through the approach to maximum packing.

2. Experimental Materials and methods

2.1. Materials

Pyromellitic dianhydride (PMDA, >98%, Tokyo Chemical Industry, Japan), 4,4'-oxydianiline (ODA, >98%, TCI), and hexagonal boron nitride (h-BN, 99.5%, mean particle diameter ~ 70 nm, MK Impex Corp., Canada) were used as received without further purification. N-Methyl-2-pyrrolidone (NMP, 99.5%) and acetone (99.5%) were purchased from Duksan Pure Chemicals, Korea.

2.2. PAA/h-BN Dispersion-Synthesis dispersion synthesis

ODA (10 mmol, 2.0024 g) was dissolved in NMP (23.707 g, 15 wt% total solids) at room temperature: h-BN (0, 10, 30, 50, or 70 wt% relative to total PAA+BN mass) was dispersed and h-BN was dispersed in this solution by probe sonication (30 min) to minimize agglomeration. PMDA (10 mmol, 2.1812 g) was added portionwise with continuous stirring at room temperature for 24 h, yielding a viscous PAApoly(amic acid) (PAA)/h-BN composite solution at equimolar PMDA:ODA ratio. [editorial comment removed] h-BN was added at 0, 10, 30, 50, or 70 wt% of the total PAA + h-BN solids while the PAA concentration in NMP was held at 15 wt% for all compositions, so that the phase-separation thermodynamics of the polymer solution were identical across the series and h-BN loading was the only compositional variable.

2.3. NIPS Film-Fabrication film fabrication and Thermal-Imidization thermal imidization

PAA/h-BN solutions were spin-coated onto soda-lime glass substrates (1000 rpm, 10 s)—[editorial comment removed]; the high viscosity of the 15 wt% PAA solution and the short spin time leave a wet film of several hundred micrometres, which after NIPS and imidization yields free-standing porous films of 52–172 μm (Figure 1). Films were immediately immersed in a 20 mL acetone non-solvent bath at room temperature for 2 h to induce phase separation. The solidified PAA/h-BN green films were thermally imidized at 200 °C for 2 h followed

by 350 °C for 1 h under ambient atmosphere, then peeled from the glass to yield free-standing porous films designated NP (~~Non-Pressed~~non-pressed). Complete imidization was confirmed by FT-IR (Figure S8): characteristic imide C=O stretching at 1780 cm⁻¹ (asymmetric) and 1720 cm⁻¹ (symmetric), C–N stretching at 1380 cm⁻¹, and complete disappearance of the PAA amide carbonyl band at ~1660 cm⁻¹.

2.4. ~~Hot-Press Densification~~Hot-press densification

A subset of NP films was densified by uniaxial hot-pressing between polished steel plates in a hydraulic press to produce HP (~~Hot-Pressed~~hot-pressed) films, adapting the hot-pressing protocol established in our previous work on NIPS-derived porous PI composite films [28]: temperature 250 °C, applied pressure 400 MPa, dwell time 30 min~~[editorial comment removed]~~. Hot-pressing partially collapses the NIPS pore network, increasing bulk density and tensile strength while reducing porosity. ~~Since-Because~~ the pressing temperature (250 °C) lies ~~far-well~~ below the glass transition temperature of PMDA–ODA PI (> 400 °C [28]), densification proceeds by ~~thermally-assisted~~thermally assisted plastic collapse (buckling and yielding) of the glassy pore walls rather than by viscous flow~~above T_g~~ . ~~This is~~ consistent with the partial, rather than complete, porosity reduction observed and with the resistance of the rigid h-BN network to compaction ~~discussed in~~ (Section 4.1). NP and HP films at identical composition were characterized side-by-side for all properties.

2.5. Characterization

Tensile properties were measured by universal testing machine (UTM, 5 mm min⁻¹ crosshead speed; rectangular specimens: width 10 mm, gauge length 50 mm, thickness measured by micrometer; ASTM D882). Dielectric constant was measured at 100 kHz ~~by-with an~~ LCR meter (~~in-house~~)~~[model, manufacturer];~~ parallel-plate contact electrodes, [diameter] mm) on films [with/without] sputtered electrodes; in addition, an independent verification measurement of the HP 30 wt% film was performed at KOPTRI (Korea Polymer Testing & Research Institute) according to ASTM D150 (100 kHz to 13 MHz; Agilent LCR; G10 electrode; ~~23(2) °C, 45(5) %~~ 23 ± 2 °C, 45 ± 5 % RH); the inter-laboratory comparison is discussed in ~~Sections 3.4 and~~ Section 4.2. Thermal diffusivity (α , laser flash) and specific heat capacity (C_p , DSC, ~~[editorial comment removed]~~ASTM E1269; 5 °C min⁻¹; 0–300 °C) were measured at Korea University of Technology and Education (KOREATECH); bulk density (ρ) was determined from film mass and optically measured dimensions; thermal conductivity was calculated as $\lambda = \alpha\rho C_p$. CTE was measured by TMA (Q400 EM, TA Instruments; tension mode; 10 °C min⁻¹; N₂ 100 mL min⁻¹; preload 0.05 N; 100–200 °C range; Kyung Hee University). ~~Porosity-Open porosity~~ was determined by mercury intrusion porosimetry (PM33GT, Quantachrome; Cooperative Equipment Center)~~[editorial comment removed]~~, [institution]. Total porosity was calculated from the bulk density as $\phi_{\text{total}} = 1 - \rho/\rho_{\text{solid}}$, with ρ_{solid} from the PI and h-BN densities and the solid-phase composition (Section 3.1); the two quantities are compared in Section 4.1. Microstructure and elemental distribution were examined by FE-SEM/EDS (JEOL-7800F and JEOL-7001F). FT-IR spectra were recorded on a FT/IR-460 Plus spectrometer (Jasco, Tokyo, Japan; 600–4000 cm⁻¹).

All properties were measured on one specimen per composition and processing state ($n = 1$) unless stated otherwise, so the reported values carry the measurement uncertainty only. Representative uncertainties are $\pm 5\%$ in α , $\pm 3\%$ in C_p , and $\pm 8\%$ in ρ (limited by thickness measurement), giving a propagated uncertainty of about $\pm 10\%$ in $\lambda = \alpha \rho C_p$; $\pm [\text{value}]$ in ε_r ; and $\pm [\text{value}]$ ppm K $^{-1}$ in CTE. These uncertainties bound the interpretation of individual residuals in Sections 4.2 and 4.3.

3. Modeling ~~Framework~~framework

3.1. Sequential ~~3-Phase-three-phase~~ Lewis–Nielsen ~~Model~~model

Each PI/h-BN porous film is modeled as a three-component system: PI matrix (property k_m), h-BN filler (property k_f), and air pores (property $k_{\text{pore}} \approx k_{\text{air}}$). We note that the term “three-phase Lewis–Nielsen model” has previously been used for a matrix–filler–interphase construction, in which the third phase is the interfacial polymer layer surrounding the nanofiller [29]; the present formulation is distinct, with the third phase being the NIPS-generated air pore network. The sequential mixing strategy [25] proceeds in two steps. In ~~Step~~step 1, the PI matrix and h-BN filler are combined to yield the dense (non-porous) composite property k_{dense} :

$$\frac{k_{\text{dense}}}{k_m} = \frac{1 + A_{\text{BN}} B \phi_{\text{BN}}}{1 - B \psi \phi_{\text{BN}}}, \quad (1)$$

where

$$B = \frac{k_f/k_m - 1}{k_f/k_m + A_{\text{BN}}}, \quad \psi = 1 + \frac{1 - \phi_{\text{max}}}{\phi_{\text{max}}^2} \phi_{\text{BN}}. \quad (2)$$

In ~~Step~~step 2, air pores are introduced into the dense composite:

$$\frac{k_{\text{final}}}{k_{\text{dense}}} = \frac{1 + A_{\text{pore}} B_p \phi_p}{1 - B_p \psi_p \phi_p}. \quad (3)$$

Although the Lewis–Nielsen model was originally formulated for elastic moduli and thermal conductivity [23, 24], its application to the dielectric constant is formally exact: permittivity and thermal conductivity are both scalar transport properties governed by the same Laplace-type field equation, $\nabla \cdot (k \nabla u) = 0$, so all effective-medium formalisms transfer between them under the substitution ~~$k \leftrightarrow \varepsilon_r$~~ $k \leftrightarrow \varepsilon$. Indeed, for spherical inclusions in the dilute limit the LN expression reduces to the Maxwell–Garnett form, of which it is a generalization through the shape factor A and the packing term ψ . The ~~decisive~~ advantage of LN over the standard dielectric mixing rules (Lichtenecker, Maxwell–Garnett, Bruggeman) for the present system is twofold: it describes both ε_r and λ with one shared microstructural parameterization, and it is the only formalism among these that carries an explicit pore-geometry parameter (A_{pore}) and a packing-limit divergence—, the two features on which the central conclusions of this work rest. A quantitative benchmark (Supplementary [Material](#), Section S3.5) confirms that the calibrated LN model outperforms all standard zero-parameter mixing rules, with the largest margin (RMSE 0.26 vs. 0.55–0.90 ~~) in the moderate-porosity~~ ~~over the 30–70 wt% films) in the moderate-porosity~~ range.

The same formalism applies identically to dielectric constant ε_r and thermal conductivity λ . The weight fraction of h-BN was converted to volume fraction in the solid (PI + ~~BN~~h-BN) phase using $\rho_{\text{PI}} = 1.42 \text{ g cm}^{-3}$ and $\rho_{\text{BN}} = 2.29 \text{ g cm}^{-3}$. ~~Measured~~The measured open porosity from mercury porosimetry was used directly intrusion was used as ϕ_p in Eq. 3; the relation between open and total porosity is examined in Sections 4.1 and 4.2. Full derivation, parameter tables, and the complete Python implementation are provided in Supplementary Material (Sections S2–S5).

3.2. Physical ~~Basis~~ basis of the Einstein ~~Coefficient~~ coefficient A

The Einstein coefficient A is the central microstructure parameter in the Lewis–Nielsen model, originating from the classical treatment of field or flux distortion around a single inclusion in an infinite matrix [23, 24]. Two distinct A values are required in the present three-phase system: A_{BN} for the h-BN filler and A_{pore} for the air pore network.

For platelet fillers, the theoretical A depends on the aspect ratio $\alpha = D/t$ and orientation relative to the applied field. For the h-BN particles used here ($\sim 70 \text{ nm}$ mean diameter), the nominal bulk aspect ratio would give $A_{\text{rand}} \sim 4\text{--}14$. However, three physical factors systematically reduce the effective aspect ratio: (i) probe sonication introduces edge fragmentation and partial delamination, reducing effective α toward 5–7; (ii) 70 nm ~~BN~~h-BN particles exhibit a high tendency to form face-to-face stacks, creating more equiaxed effective inclusions; and (iii) rapid NIPS solvent exchange freezes particles in random orientations, preventing in-plane alignment. For randomly oriented platelets of effective $\alpha \approx 5\text{--}7$, $A_{\text{rand}} \approx 4\text{--}6$ and $\phi_{\text{max}} \approx 0.55\text{--}0.65$ [23, 24, 25]; these values are adopted for the thermal model ($A = 5.5$, $\phi_{\text{max}} = 0.60$ fixed; Section 3.4). For the dielectric model, the fitted $A_{\text{BN}} = 0.68$ falls below the isolated-sphere value ($A = 1.5$); however, the sensitivity analysis (Section 4.6, Figure S1a) demonstrates that ε_r predictions are nearly insensitive to A_{BN} across 0.3–6.0—; the pore network (~~Step~~step 2, Eq. 3) dominates the dielectric response, so the dielectric A_{BN} is only weakly determined by the data and its precise value carries little physical weight.

For the NIPS-generated pore network, A captures how effectively the dispersed pore phase shields the matrix from the applied field. For isolated spherical pores, $A = 1.5$. For the bicontinuous sponge morphology produced by NIPS, the PI skeleton forms continuous conduction paths through the pore space—meaning, so the field disruption per unit porosity is greatly reduced. The fitted $A_{\text{pore}} = 0.08$ quantifies this geometry: at $\phi_p = 0.49$ (NP 30 wt% film), isolated spherical pores ($A = 1.5$) would ~~predict~~give $\varepsilon_r \approx 2.22$, while ~~whereas~~ $A_{\text{pore}} = 0.08$ ~~predicts~~gives $\varepsilon_r \approx 1.71$ —a; the difference of 0.51 ~~units that directly~~ reflects the bicontinuous sponge architecture. This A_{pore} value constitutes a quantitative microstructure fingerprint distinguishing NIPS sponge pores from spherical Pickering emulsion (HIPPE) pores [21], for which $A \sim 1.2\text{--}1.5$ would be expected.

3.3. CTE and ~~Mechanical Models~~mechanical models

CTE was predicted using the Turner model [26]: $\text{CTE}_c = (\sum_i \phi_i E_i \alpha_i) / (\sum_i \phi_i E_i)$, with $E_{\text{BN}} = 200 \text{ GPa}$ and $\alpha_{\text{BN}} = 1.5 \text{ ppm/K}$ ~~$\alpha_{\text{BN}} = 1.5 \text{ ppm K}^{-1}$~~ [16]. For α_{PI} , the ~~measured neat PMDA–ODA porous PI value~~

(HP sample #1, TMA) of 41.92 ppm/K was used — value measured on the neat porous PI HP film (TMA), 41.92 ppm K^{-1} was used rather than the commonly cited dense PI — dense-PI literature value of $\sim 60 \text{ ppm/K}$ — 60 ppm K^{-1} because the porous sponge architecture modifies the effective thermal expansion of the PI phase. $E_{\text{PI}} = 3.5 \text{ GPa}$ was taken from literature for PMDA–ODA PI [5]. For comparison, the linear rule-of-mixtures (ROM) prediction is also shown. Notably, the Turner model substantially underpredicts and the ROM overpredicts the experimental CTE at 10–70 wt% BN-h-BN. This is attributed to the porous sponge architecture violating the continuous load-path assumption of the Turner model: h-BN platelets embedded in discontinuous thin PI ligaments cannot exert the same macroscopic elastic constraint as in a fully dense co-continuous composite. The experimental CTE values therefore lie between the upper bound (ROM, no constraint) and the lower bound (Turner, full constraint), a physically reasonable outcome for a partially load-bearing porous skeleton. Tensile strength was modeled via the LN composite modulus prediction combined with the Gibson–Ashby open-cell foam scaling ($\sigma \propto E_{\text{dense}}(1 - \phi_p)^2$) [27].

3.4. Parameter ~~Estimation Protocol~~ estimation and Model Validation model validation

Dielectric model. ~~Four~~ We optimized four parameters (A_{BN} , $\phi_{\text{max,BN}}$, A_{pore} , $\phi_{\text{max,pore}}$) ~~were simultaneously optimized~~ simultaneously to minimize the sum of squared residuals over all ~~10-ten~~ ϵ_r measurements (~~5-five~~ compositions $\times$ ~~2-two~~ conditions) using multi-start Nelder–Mead optimization (16 initial conditions; Eq. S8). The packing parameters converge to the upper boundary (0.99) and are effectively inactive: a sensitivity scan (Supplementary–Table S11) shows that fixing $\phi_{\text{max,BN}}$ anywhere in 0.50–0.99 changes the RMSE by only 0.03. The dielectric model is therefore governed by 2–3 active parameters fitted to ~~10-ten~~ points.

Thermal model. When all three thermal parameters ($\lambda_{\text{BN,eff}}$, $A_{\text{BN},\lambda}$, $\phi_{\text{max,BN},\lambda}$) are fitted without constraints, the optimizer converges to a degenerate solution ($A \rightarrow 700$, $\phi_{\text{max}} \rightarrow 0.10$) because A and ϕ_{max} are strongly correlated in Eq. 1 and the data cannot separate them; the resulting parameters are physically meaningless even though the fit quality is good (Supplementary–Section S4.2). To eliminate this degeneracy, ~~we reduced~~ the thermal model ~~was deliberately reduced~~ to a single fitted parameter: $A_{\text{BN},\lambda}$ was fixed at 5.5 and $\phi_{\text{max,BN},\lambda}$ at 0.60 —, ~~the~~ literature values for randomly oriented platelets with effective aspect ratio $\alpha \approx 5$ –7 [23, 24, 25] —, ~~and~~ consistent with the sonication-modified particle geometry discussed in Section 3.2 —, and only $\lambda_{\text{BN,eff}}$ was fitted, yielding ~~$6.81 \text{ W m}^{-1} \text{ K}^{-1}$~~ $\lambda_{\text{BN,eff}} = 6.81 \text{ W m}^{-1} \text{ K}^{-1}$ with RMSE (0.027 NP, ~~0.030 HPW/mK~~ $0.030 \text{ W m}^{-1} \text{ K}^{-1}$ HP) statistically indistinguishable from the unconstrained three-parameter fit (0.027, 0.023). The thermal model thus has a 7:1 data-to-parameter ratio, and all reported parameters are physically interpretable. Notably, with $\phi_{\text{max}} = 0.60$, the 70 wt% composition ($\phi_{\text{BN}} = 0.591$) sits at 99% of maximum packing, so the packing-interaction term ψ in Eq. 1 ~~naturally~~ generates the sharp conductivity upturn observed ~~experimentally~~ at this loading —; ~~this is~~ the LN representation of ~~approaching the approach to~~ the particle-contact (percolation) limit.

Validation. ~~An independent KOPTRI measurement of the HP 30 wt% film gives $\epsilon_r = 1.610$, compared with 1.864 in-house (–13.6%–~~ The calibrated dielectric model is compared with an independent inter-laboratory

deviation) and a model prediction of 1.376 at the measured porosity of 67.2%. All three values disagree pairwise at the 14% level — and all of the disagreement traces to the same composition flagged for the porosity anomaly measurement in Section 4.1: if the HP 30 wt% porosity lay in the 40–45% range expected from the monotonic NP→HP densification trend (rather than the anomalous measured 67.2%), the model would predict $\varepsilon_r = 1.79$ –1.92, in agreement with the in-house measurement. The inter-laboratory comparison therefore reinforces, rather than undermines, the conclusion that the 30 wt% HP porosity measurement requires re-examination. Model credibility instead 4.2. Beyond that comparison, model credibility rests on three pillars: (i) the parsimony of the calibration (2–3 active dielectric parameters, 1-one thermal parameter); (ii) the shape-factor sensitivity analysis (Section 4.6) demonstrating that conclusions are robust to the weakly determined parameters; and (iii) the agreement of the model λ envelope with the experimental 70 wt% data (Section 4.8, Figure ??e4d).

Table 1: ~~3-phase~~ Three-phase Lewis–Nielsen parameters: estimation protocol and physical interpretation.

Parameter Symbol Value Status Physical interpretation h-BN shape factor (ε_r) 0.68 Fitted (weakly det.) ε_r is pore-dominated; carries little weight (Figure S1a) h-BN max. packing ($\phi_{\max, \text{BN}}$) 0.99 Boundary (inactive) RMSE changes ≤ 0.03 over 0.50–0.99 (Table S11) Pore shape factor 0.08 Fitted Bicontinuous NIPS sponge; cf. $A = 1.5$ for isolated spheres Pore max. packing $\phi_{\max, \text{pore}}$ 0.99 Boundary NIPS sponge supports up to 91% open porosity h-BN shape factor (λ) $A_{\text{BN}, \lambda}$ 5.5 Fixed (literature) Randomly oriented platelet, effective $\alpha \approx 5$ –7 h-BN max. packing ($\phi_{\max, \text{BN}, \lambda}$) 0.60 Fixed (literature) Random platelet packing; $\phi_{\text{BN}}(70 \text{ wt}\%) = 0.59 \rightarrow 99\%$ of Eff. BN conductivity 6.81 W/m K Fitted (single) $\approx 7\times$ below bulk (~ 50); Kapitza resistance at 70 nm

Parameter	Symbol	Value	Status	Physical interpretation
h-BN shape factor (ε_r)	A_{BN}	0.68	Fitted (weakly det.)	ε_r is pore-dominated; A_{BN} carries little weight (Figure S1a)
h-BN max. packing (ε_r)	$\phi_{\max, \text{BN}}$	0.99	Boundary (inactive)	RMSE changes ≤ 0.03 over 0.50–0.99 (Table S11)
Pore shape factor	A_{pore}	0.08	Fitted	Bicontinuous NIPS sponge; cf. $A = 1.5$ for isolated spheres
Pore max. packing	$\phi_{\max, \text{pore}}$	0.99	Boundary	NIPS sponge supports up to 91% open porosity
h-BN shape factor (λ)	$A_{\text{BN}, \lambda}$	5.5	Fixed (literature)	Randomly oriented platelet, effective $\alpha \approx 5$ –7
h-BN max. packing (λ)	$\phi_{\max, \text{BN}, \lambda}$	0.60	Fixed (literature)	Random platelet packing; $\phi_{\text{BN}}(70 \text{ wt}\%) = 0.59 \rightarrow 99\%$ of $\phi_{\max}$
Eff. h-BN conductivity	$\lambda_{\text{BN}, \text{eff}}$	6.81 W m ⁻¹ K ⁻¹	Fitted (single)	Well below the bulk through-plane value; Kapitza resistance at 70 nm

4. Results and Discussion

4.1. Film ~~Morphology~~: morphology in the NP vs. and HP Processing States

Two distinct film states result from NIPS fabrication and optional hot-press densification. ~~Non-Pressed~~ Non-pressed (NP) films retain the full porous sponge architecture generated during NIPS: acetone non-solvent exchange produces a bicontinuous pore network with characteristic pore sizes of 1–10 μm confirmed by FE-SEM. Measured open porosities range from 48.8% (30 wt% h-BN) to 91.4% (10 wt% h-BN). The ~~anomalously~~ high porosity at 10 wt% ~~BN is attributed to h-BN~~ may reflect a nucleation effect: ~~in which~~ h-BN nanoparticles at this loading ~~may~~ act as additional phase-separation nuclei ~~producing finer and~~ and

produce finer, more uniformly distributed pores. FE-SEM cross-sections confirm the sponge-like open-cell architecture in all NP films, with h-BN platelets embedded within the PI cell walls. Figure 1 presents the cross-sectional morphology of all ten films; [photographs and top-view micrographs are given in Figures S6 and S7](#).

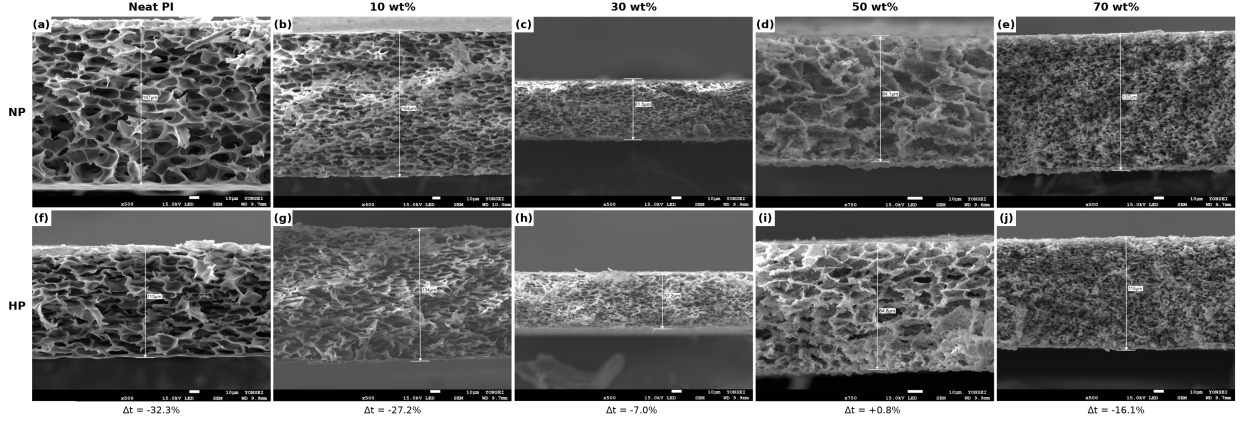


Figure 1: Cross-sectional FE-SEM micrographs of PI/h-BN films at 0–70 wt% h-BN in the non-pressed (a–e, top row) and hot-pressed (f–j, bottom row) states. Δt denotes the film-thickness change upon hot-pressing; ~~asterisks mark the 30- and 50 wt% compositions flagged for anomalous porosity behavior (see text)~~. Top-view micrographs and EDS elemental analysis are provided in ~~Supplementary Figures~~ [Figure S3–S7](#) and Table S1.

~~Hot-Pressed~~ [Hot-pressed](#) (HP) films are produced by applying uniaxial compressive force at elevated temperature to NP films. Hot-pressing ~~selectively collapses portions~~ [collapses part](#) of the open pore network, increasing bulk density and tensile strength. For most compositions, ~~HP porosities are the~~ [HP open porosity is](#) substantially lower than ~~NP the NP value~~ (43.4% vs. 75.4% for neat PI; 29.8% vs. 91.4% for 10 wt% BN). ~~However, h-BN), whereas the 30 wt% and 50 wt% HP films exhibit anomalously higher show higher intrusion porosities than their NP counterparts (67.2% vs. 48.8% and 64.6% vs. 58.1%), accompanied by near-zero film thickness reduction at.~~ [Mercury intrusion probes only the open, connected pore volume. The total porosity from the bulk density, \$\phi_{\text{total}} = 1 - \rho/\rho_{\text{solid}}\$ \(Table 3\), exceeds the intrusion value for six of the seven films for which \$\rho\$ was measured, by up to 31 percentage points \(NP 30 wt%: 79.8% vs. 48.8%\); the difference is a measure of closed porosity. Notably, \$\phi_{\text{total}}\$ decreases from NP to HP at every composition \(30 wt%: 79.8 \$\rightarrow\$ 71.6%; 50 wt% \(–0.8%\). This behavior is attributed to structural redistribution of h-BN platelets under compressive load: the rigid BN network resists overall compaction while internal pore connectivity changes increase the measurable open-channel porosity. Re-measurement of these two compositions by an independent technique \(e.g., gas pycnometry\) is recommended to resolve this anomaly. \[editorial comment removed\] To assess the sensitivity of the LN model to this uncertainty, predictions were compared using the measured anomalous HP porosities versus their expected \(lower\) values: the resulting change in predicted was \$\pm 0.12\$ at most \(for wt%: 70.6 \$\rightarrow\$ 61.3%; 70 wt%: 72.8 \$\rightarrow\$ 72.1%\), so the reversed NP/HP ordering of the intrusion values at 30 and 50 wt% HP\), which is within the overall model RMSE and does not qualitatively alter any conclusion. The inter-laboratory dielectric comparison at this same composition \(Section 3.4\) independently corroborates the anomaly reflects a change in pore](#)

connectivity, in which compaction opens previously closed pores to intrusion, rather than an increase in total pore volume. Both quantities are tabulated in Table 3, and the choice of porosity input to Eq. 3 is discussed in Section 4.2. SEM cross-sections (Figure 1) confirm progressive densification at 0, 10, and 70 wt% BN (thickness reductions of 32.3%, 27.2%, and 16.1% thickness reduction, respectively), while at 0, 10, and 70 wt% h-BN, whereas the 30 and 50 wt% films show near-zero thickness change (-7.0% and $+0.8\%$)—the microstructural counterpart of the anomalous porosity behavior, consistent with the resistance of the h-BN network to compaction at these loadings. EDS elemental analysis (Supplementary Table S1) confirms that the boron content scales systematically with nominal loading (B: 5.4 ± 8.0 to 29.3 wt% from 0–10 to 70 wt% h-BN, NP series) and is preserved upon hot-pressing, verifying composition retention through the NIPS and densification steps; the apparent B signal of 5.4 wt% in the neat film reflects the quantification limit of EDS for light elements and is treated as background.

EDS elemental mapping confirms near-quantitative h-BN incorporation across all compositions at >10 wt% h-BN, with measured boron weight fractions within ± 5 wt% (absolute) of the theoretical values. Spot-to-spot variance (<1 wt% B at 70 wt% BN/h-BN) confirms homogeneous h-BN dispersion achieved by the sonication-assisted NIPS protocol.

4.2. Dielectric Constant: Experiment and 3-Phase LN Model constant

Figure ??-2a presents the measured dielectric constants and 3-phase LN model—the three-phase LN predictions for all 10 samples; the corresponding parity plot is given in Figure S2. All films exhibit ϵ_r far below dense PI (~ 3.5), confirming that NIPS-generated porosity dominates the dielectric response. NP films achieve reach $\epsilon_r = 1.28$ – 1.76 (100 kHz); and HP films span $\epsilon_r = 1.62$ – 2.18 . The monotonic decrease of ϵ_{NP} with increasing BN loading ($1.76 \rightarrow 1.28$) reflects increasing BN solid content combined with generally high porosity. Hot-pressing uniformly raises ϵ_r by $\Delta\epsilon_r = 0.18$ – 0.42 at every composition through pore collapse. In the NP series ϵ_r decreases monotonically with h-BN content ($1.76 \rightarrow 1.28$); because ϵ_{BN} (~ 6) exceeds ϵ_{PI} (~ 3.5) and the open porosity does not increase over this range, no mixing rule reproduces this trend (Section S3.5), and the model predictions evaluated at each film’s open porosity follow the porosity input rather than the composition (Figure 2a).

The LN model returns RMSE = 0.43 (NP) and 0.36 (HP); [editorial comment removed], which exceeds the standard deviation of the measured ϵ_r across compositions (0.19 NP, 0.18 HP); the coefficient of determination is therefore negative ($R^2 = -3.9$ NP, -3.0 HP; -1.6 for the 30–70 wt% subset). The calibrated model does not predict ϵ_r for individual films better than the data mean, and we use it accordingly: as a mechanistic description of how porosity and pore geometry enter ϵ_r , and as a generator of design maps whose absolute levels carry an uncertainty of the order of the RMSE. Part of the residual is attributable to the porosity input itself. The fits use the open porosity as ϕ_p in Eq. 3, the quantity available for all ten films; because ϵ_r and λ respond to the total air fraction irrespective of its connectivity, ϕ_{total} is the physically appropriate input, but it is available only for the seven films with measured density and differs from the open porosity by up to 31 percentage points (Section 4.1). The largest residuals occur at 0 and

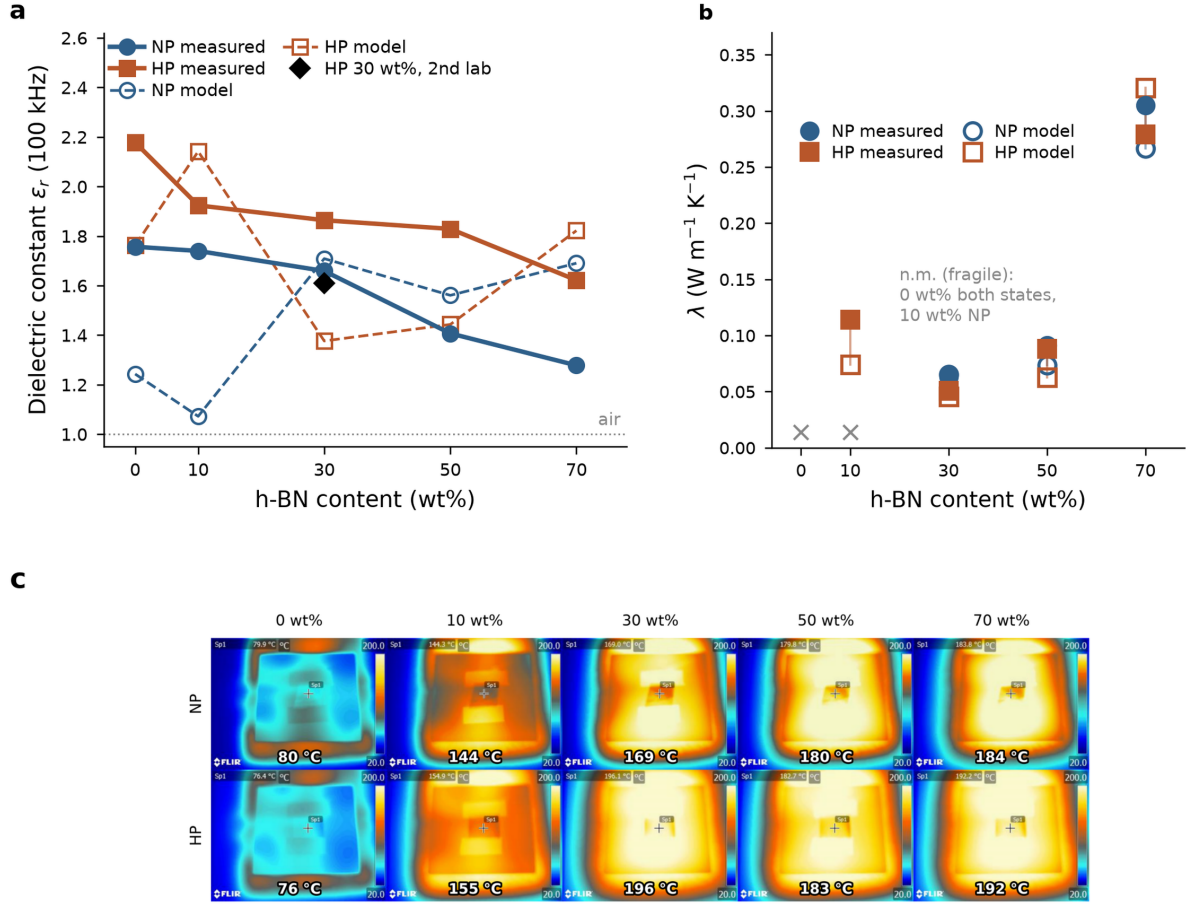


Figure 2: Measured dielectric and thermal properties of PI/h-BN films ($n = 1$ per condition). (a) Dielectric constant (at 100 kHz) vs. h-BN content for non-pressed (NP) and hot-pressed (HP) PI/h-BN composite films. Filled: filled symbols: measured values ($n = 1$ specimen per condition). Open: open symbols joined by dashed lines: 3-phase Lewis-Nielsen three-phase LN predictions evaluated at each film's own measured open porosity. Black: black diamond: independent inter-laboratory measurement of the HP 30 wt% film (KOPTRI, ASTM D150), not used in fitting. (b) Parity plot of predicted Thermal conductivity vs. measured dielectric constant for all 10 samples h-BN content; open symbols, with $\pm 10\%$ and $\pm 20\%$ bands. The panel title reports the model RMSE alongside the standard deviation of the data itself single-parameter LN predictions ($\lambda_{\text{BN,eff}}$ fitted; because RMSE exceeds $\text{SDA} = 5.5$, $\phi_{\text{max}} = 0.60$ fixed), joined to the calibrated model does measured value by a tie-line; crosses mark compositions that could not outperform the data mean be measured (see Section 4.2 film fragility). (c) Infrared thermography of all ten films on a 200 °C stage after [X] min equilibration; annotated values are uncorrected radiometric spot temperatures and are comparative, not absolute. Parity plot and porosity-sensitivity curves are given in Figures S2 and S3.

10, and 70 wt% ~~BN in the NP series~~ (model underprediction of 29–38% ~~at 0–10 wt% and overprediction of 32% at 70 wt%~~, Table 2).

Two physical mechanisms explain ~~this the~~ systematic underprediction at low ~~BN-h-BN~~ loading. First, at these compositions, the NP films exhibit extremely high porosities (75.4% and 91.4%, respectively). At $\phi_p > 0.75$, the sponge pore network transitions from a regime of moderate connectivity (where the mean-field $A_{\text{pore}} = 0.08$ provides adequate description) to a near-fully-open framework where the PI skeleton itself becomes discontinuous. In this regime, the effective dielectric constant is dominated by long-range pore connectivity and wall-thinning effects that are outside the scope of a local mean-field mixing rule. Second, at 10 wt% ~~BN-h-BN~~ with 91.4% porosity, the ~~actual BN-h-BN~~ volume fraction in the film is only $\phi_{\text{BN, film}} = 0.006$ ~~—effectively negligible—meaning—, so~~ the dielectric response should approach that of ~~pure a~~ PI foam. For the limiting inverted topology (isolated PI inclusions in air), the dilute Maxwell–Garnett estimate $\epsilon_r \approx 1 + 3\phi_s(\epsilon_{\text{PI}} - 1)/(\epsilon_{\text{PI}} + 2)$ at solid fraction $\phi_s \approx 0.087$ gives $\epsilon_r \approx 1.12$ ~~—, also far below the measured 1.74.~~ The NIPS film at this extreme porosity likely exhibits a partially inverted topology that neither the Maxwell–Garnett nor the Lewis–Nielsen mean-field picture captures: the measured value indicates that the real PI skeleton remains substantially more connected than either idealized topology assumes. These limitations are inherent to mean-field approaches and do not affect conclusions at compositions with moderate porosity (30–70 wt% ~~BN range h-BN~~), where ~~restricting the RMSE evaluation to 30–70~~ ~~the RMSE restricted to the NP films of this range is 0.25.~~

~~An independent measurement of the HP 30 wt% gives 0.25 (NP) film at KOPTRI (ASTM D150) gives $\epsilon_r = 1.610$, compared with 1.864 in-house (–13.6% inter-laboratory deviation) and a model prediction of 1.376 at the open porosity of 67.2% (Figure 2a, black diamond; Table 2). The three values disagree pairwise at the 14% level. The inter-laboratory difference is an estimate of the reproducibility of ϵ_r measurements on porous films with contact electrodes: an air gap between a rough porous surface and the electrode acts as a series capacitance that lowers the apparent ϵ_r by an amount that depends on surface texture, and therefore on composition and processing state. This effect is a plausible experimental contributor to the negative R^2 and to the composition trend of the NP series noted above, and re-measurement with sputtered or painted electrodes would resolve it. The model residual at 30 wt% HP (–26%) exceeds the inter-laboratory spread and is therefore a model limitation rather than a measurement artefact.~~

4.3. Thermal ~~Conductivity: Experiment and Lewis–Nielsen Fit~~ conductivity

Figure ~~??a–2b~~ shows thermal conductivity results, and Table 3 lists the underlying laser-flash data (bulk density, specific heat, and thermal diffusivity) ~~together with the open and total porosities.~~ Measurements were unavailable for the neat films in both states and for NP 10 wt% due to film fragility at these high-porosity compositions (up to 91.4%), leaving seven measured data points. For all available data, the single-parameter LN calibration reproduces the trends well ($\text{RMSE}_{\text{NP}} = 0.027$, $\text{RMSE}_{\text{HP}} = 0.030 \text{ W m}^{-1} \text{ K}^{-1}$): near-constant λ at 30–50 wt% ~~BN-h-BN~~ and a sharp increase at 70 wt%. Within the model, the 70 wt% upturn arises ~~naturally~~ from the packing-interaction term ψ in Eq. 1: at $\phi_{\text{BN}} = 0.591 \approx 0.99 \phi_{\text{max}}$, the

Table 2: Experimental vs. predicted dielectric constants (~~3-phase~~ three-phase Lewis–Nielsen model). KOPTRI: independent inter-laboratory measurement of the HP 30 wt% film (ASTM D150; not used in fitting).

h-BN (wt%)	$\epsilon_{\text{NP}}^{\text{LN}}$ (Exp.)	$\epsilon_{\text{NP}}^{\text{LN}}$ (LN)	Δ (%)	$\epsilon_{\text{HP}}^{\text{LN}}$ (Exp.)	$\epsilon_{\text{HP}}^{\text{LN}}$ (LN)	Δ (%)	KOPTRI (HP)
0	1.757	1.242	−29.3	2.178	1.760	−19.2	—
10	1.740	1.073	−38.4	1.924	2.139	+11.2	—
30	1.659	1.707	+2.9	1.864	1.376	−26.2	1.610*
50	1.407	1.560	+10.8	1.829	1.442	−21.2	—
70	1.278	1.689	+32.1	1.622	1.819	+12.2	—

$\Delta = (\epsilon_{\text{LN}} - \epsilon_{\text{Exp}}) / \epsilon_{\text{Exp}} \times 100\%$ $\Delta = (\epsilon_{\text{LN}} - \epsilon_{\text{Exp}}) / \epsilon_{\text{Exp}} \times 100\%$. LN predictions are evaluated at each film's open porosity. *Inter-laboratory deviation from the in-house HP value (1.864) is −13.6%; the model prediction at the measured (anomalous) open porosity of 67.2% is 1.376. All three values converge if the porosity lies in the expected 40–45% range —see (Section 3.4.2).

composite approaches the maximum platelet packing, which is the Lewis–Nielsen representation of imminent particle–particle contact (percolation).

(a) Thermal conductivity vs. h-BN content: measured laser flash values (filled symbols, $n = 1$ per condition) and 3-phase LN predictions evaluated at each film's own measured porosity (open symbols), joined by a vertical tie-line so the per-sample deviation is directly readable; the single-parameter thermal calibration is used throughout (fitted; $A = 5.5$, $\phi_{\text{max}} = 0.60$ fixed). Crosses mark compositions for which could not be measured (film fragility): neat PI in both states and NP 10 wt%. (b) Porosity sensitivity at fixed 50 wt% h-BN: (left axis, solid) and (right axis, dashed) both rise as porosity falls, so porosity alone cannot decouple the two properties; circles and squares mark the measured NP and HP 50 wt% films. (c) Infrared thermography of all ten films on a 200 °C stage, labeled by composition (columns) and processing state (rows); the annotated value is the Sp1 spot temperature. [editorial comment removed]

Table 3: Thermophysical Porosity and thermophysical data underlying the thermal-conductivity determination ($\lambda = \alpha \rho C_p$; laser flash at 25 °C). ϕ_{open} : mercury intrusion; $\phi_{\text{total}} = 1 - \rho / \rho_{\text{solid}}$ with $\rho_{\text{PI}} = 1.42$ and $\rho_{\text{BN}} = 2.29 \text{ g cm}^{-3}$.

h-BN (wt%)	State	ϕ_{open} (%)	ϕ_{total} (%)	ρ (g cm ^{−3})	C_p (J g ^{−1} K ^{−1})	α (mm ² s ^{−1})	λ (W m ^{−1} K ^{−1})
0	NP	75.4	—	—	—	—	—
0	HP	43.4	—	—	—	—	—
10	NP	91.4	—	—	—	—	—
10	HP	29.8	40.4	0.879	0.903	0.143	0.114
30	NP	48.8	79.8	0.323	0.853	0.230	0.063
30	HP	67.2	71.6	0.455	0.863	0.131	0.051
50	NP	58.1	70.6	0.515	0.875	0.202	0.091
50	HP	64.6	61.3	0.678	0.885	0.147	0.088
70	NP	54.2	72.8	0.526	0.838	0.693	0.305
70	HP	48.8	72.1	0.539	0.846	0.613	0.279

Neat PI (both states) and NP 10 wt% could not be measured by laser flash (film fragility), and ρ was not determined for these films. The LN fits use ϕ_{open} as the porosity input (Section 4.2).

The fitted $\lambda_{\text{BN,eff}} = 6.81 \text{ W m}^{-1} \text{ K}^{-1}$ is approximately 7× lower than the bulk lies well below the through-plane h-BN conductivity conductivity of bulk h-BN ($\sim 50 \text{ W m}^{-1} \text{ K}^{-1}$ [editorial comment removed]), attributable [value] $\text{W m}^{-1} \text{ K}^{-1}$ [14]), which we attribute primarily to Kapitza thermal interface resistance (TIR) at the h-BN–PI interface. At 70 nm particle diameter, the surface-area-to-volume ratio is approxi-

mately $30 \text{ m}^2 \text{ g}^{-1}$, causing interfacial phonon scattering to limit heat transfer through individual particles—; this is consistent in order of magnitude with molecular-dynamics-based interfacial conductance estimates for BN-polymer-h-BN-polymer interfaces ($G \sim 10\text{--}100 \text{ MW/m}^2 \text{ K}$ $100 \text{ MW m}^{-2} \text{ K}^{-1}$), which at 70 nm correspond to effective particle conductivities of order $1\text{--}7 \text{ W/mK}$ $1\text{--}7 \text{ W m}^{-1} \text{ K}^{-1}$ [30]. Two residual features carry physical information, with the caveat that both are comparable to the $\sim 10\%$ propagated uncertainty in λ (Section 2.5). First, the NP 70 wt% film exceeds the model prediction (0.305 vs. $0.263 \text{ W m}^{-1} \text{ K}^{-1}$, $+16\%$), consistent with incipient BN-BN contact percolation-h-BN-h-BN contact beyond the mean-field description. Second—and experimentally striking—, hot-pressing at 70 wt% reduces λ ($0.305 \rightarrow 0.279 \text{ W m}^{-1} \text{ K}^{-1}$) despite lowering porosity ($54.2\% \rightarrow 48.8\%$) while the open porosity decreases ($54.2 \rightarrow 48.8\%$) and the total porosity is nearly unchanged ($72.8 \rightarrow 72.1\%$), opposite to the model trend (which predicts an increase to 0.315): this indicates that compaction partially disrupts the BN-BN-h-BN-h-BN contact network formed during NIPS, sacrificing percolative pathways even as the solid fraction increases. Porosity sensitivity at fixed composition (Figure ??bS3) shows that both properties remain strongly porosity-coupled at 50 wt%: over the experimentally accessible 30–65% porosity range, ε_r spans 1.43–2.31 while λ spans $0.06\text{--}0.16 \text{ W m}^{-1} \text{ K}^{-1}$.

4.4. Design Maps for Simultaneous Property Optimization Tensile strength and coefficient of thermal expansion

Figure 3 presents tensile strength and CTE results. Hot-pressing consistently improves tensile strength by 19–95% (Table 4), consistent with pore collapse under compressive load. The NP tensile strength is non-monotonic, with a local maximum at 30 wt% (25.4 MPa) and minima at 10 wt% (14.9 MPa) and 50 wt% (12.5 MPa), and correlates inversely with the porosities at 10 wt% (91.4%) and 50 wt% (58.1%). The LN + Gibson–Ashby model reproduces this trend qualitatively.

For CTE (Figure 3b), the Turner model (using $\alpha_{\text{PI}} = 41.92 \text{ ppm K}^{-1}$ as measured on the neat PI HP film) correctly predicts the neat-film value but substantially underpredicts CTE at all h-BN loadings (-61 to -80%). The ROM overestimates because it neglects elastic constraint entirely. The experimental CTE values ($24.96 \rightarrow 10.11 \text{ ppm K}^{-1}$ from $10 \rightarrow 70$ wt%) lie between the two model bounds, physically consistent with partial elastic constraint in a porous sponge where h-BN platelets are embedded within discontinuous PI ligaments rather than a macroscopically continuous dense matrix. This porous-architecture effect is a limitation of applying dense-composite CTE models to sponge materials and an opportunity for future model development incorporating open-cell foam mechanics. The measured CTE of 10.1 ppm K^{-1} at 70 wt% h-BN (HP) lies within the PCB substrate compatibility window ($3\text{--}17 \text{ ppm K}^{-1}$). [CTE was measured on the HP films only]

4.5. Design maps for simultaneous property optimization

Figure ??4a,b presents model-generated design maps of ε_r and λ spanning the full composition space (h-BN content 0–70 wt%; porosity 0–95%), with experimental NP and HP states overlaid and arrows indicating the NP→HP processing trajectory at each composition.

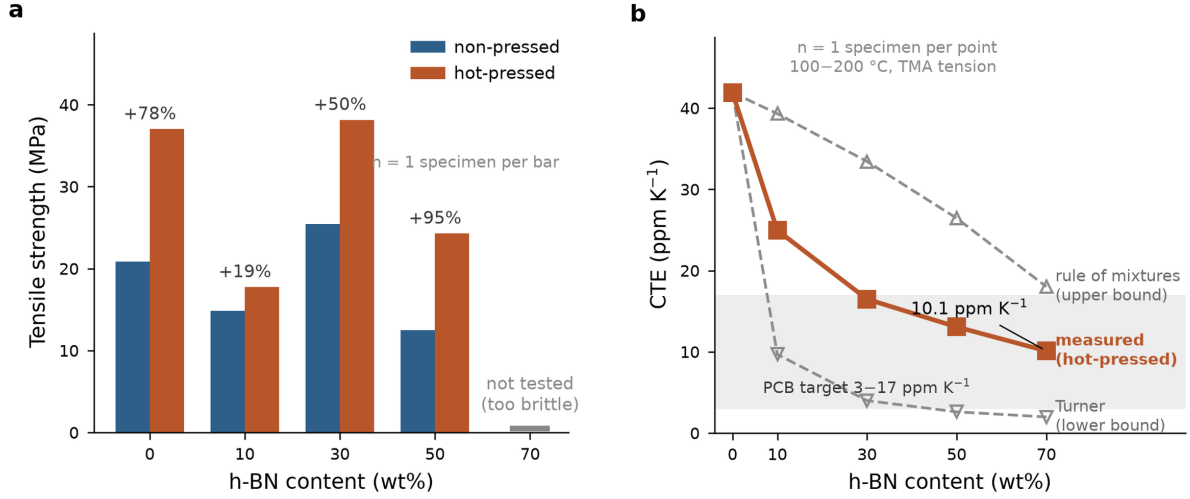


Figure 3: Mechanical and thermal-expansion behavior of PI/h-BN films. (a) Tensile strength vs. h-BN content for NP and HP films ($n = 1$ specimen per bar). Percentage values indicate the NP→HP change at each composition; the 70 wt% slot carries a baseline stub because those films were too brittle to test, not because the strength is zero. (b) CTE vs. h-BN content for HP films ($n = 1$ per point; TMA, tension mode, 100–200 °C): measured values against the Turner lower bound and the rule-of-mixtures (ROM) upper bound, both computed with $\alpha_{PI} = 41.92 \text{ ppm K}^{-1}$ (measured). The PCB substrate target zone (3–17 ppm K^{-1}) is shaded. The measured CTE lies between the two bounds because the discontinuous load-path geometry of the porous sponge gives only partial elastic constraint (Section 3.3).

Table 4: Tensile strength and CTE data for PI/h-BN composite films.

h-BN (wt%)	NP strength (MPa)	HP strength (MPa)	NP→HP (%)	HP CTE (ppm K ⁻¹)	Turner (ppm K ⁻¹)	ROM (ppm K ⁻¹)
0	20.85	37.02	+77.6	41.92	41.92	41.92
10	14.86	17.74	+19.4	24.96	9.69	39.31
30	25.43	38.15	+50.0	16.49	4.00	33.43
50	12.47	24.28	+94.7	13.06	2.61	26.45
70	—*	—*	—	10.11	1.98	18.02

*Not tested (film brittleness at 70 wt% h-BN). Turner and ROM use $\alpha_{PI} = 41.92 \text{ ppm K}^{-1}$ (measured neat porous PI, HP), $E_{PI} = 3.5 \text{ GPa}$, $E_{BN} = 200 \text{ GPa}$, $\alpha_{BN} = 1.5 \text{ ppm K}^{-1}$, solid-phase volume fractions.

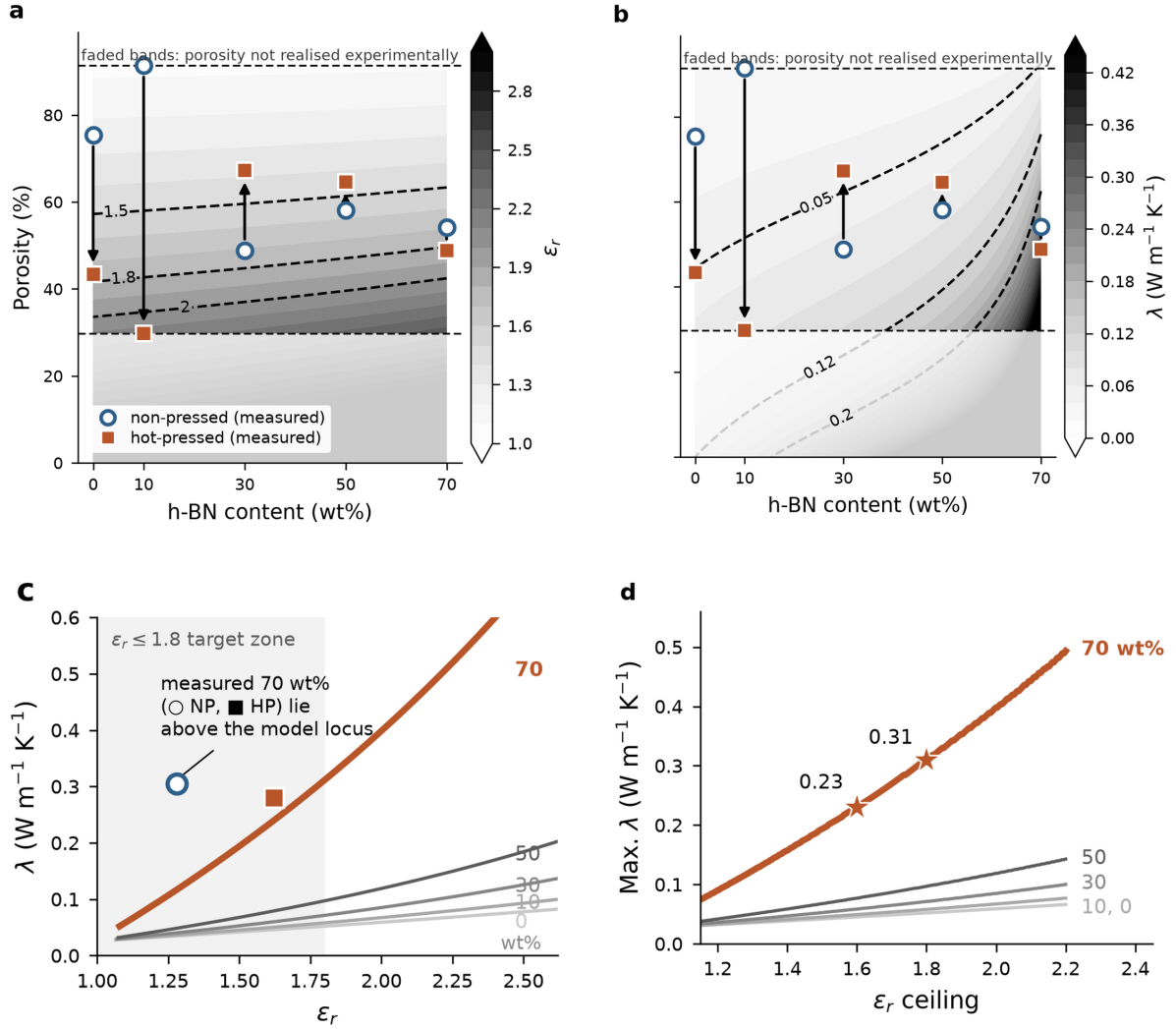


Figure 4: Design maps Model-guided design (3-phase three-phase LN model, calibration of Table 1) for. (a, b) dielectric constant Design maps of ϵ_r and (b) thermal conductivity λ (W/mK) as functions of vs. h-BN content and porosity. Measured: measured NP (open circles) and HP (filled squares) states are overlaid; arrows indicate indicating the NP → HP processing trajectory. The: faded bands above 91% 91% and below 30% 30% porosity lie outside the range realised-realized by any of film. (c) ϵ_r - λ loci for each loading with porosity swept along each curve; the ten-measured 70 wt% films, so lie above their own locus. (d) Maximum λ attainable under a given ϵ_r ceiling with porosity optimized at each point; stars mark the map there is extrapolation rather than interpolation between measured states $\epsilon_r \leq 1.6$ and $\epsilon_r \leq 1.8$ operating points at 70 wt%. The absolute levels of panels (a)-(d) inherit the dielectric uncertainty quantified in Section 4.2. Feasible-window and porosity-optimum panels are given in Figures S4 and S5.

Three actionable design principles emerge. First, achieving $\varepsilon_r < 1.5$ requires porosity above $\sim 57\%$ at low BN-h-BN loading, rising to $\sim 63\%$ at 70 wt%—comfortably, a range accessible to NIPS in both NP and moderately pressed states. Second, λ is strongly loading-gated: $\lambda > 0.20 \text{ W m}^{-1} \text{ K}^{-1}$ is unreachable below $\sim 55 \text{ wt\% BN-h-BN}$ at any practical porosity, requires porosity below $\sim 35\%$ at 60 wt%, but is available up to $\sim 62\%$ porosity at 70 wt%—; the packing-limit amplification at high loading dramatically relaxes the porosity constraint. Third, the NP→HP trajectories show that hot-pressing traverses the maps nearly vertically (porosity is the processing variable), so composition must be selected first and porosity tuned second; at 70 wt%, both the NP and HP states already lie in the simultaneously favorable region.

4.6. ~~Shape Factor Sensitivity Analysis~~ Shape-factor sensitivity

~~Supplementary~~ Figure S1 systematically examines the sensitivity of LN predictions to the h-BN shape factor A for values 0.3–6.0, corresponding to geometries from needle-like rods to highly aligned platelets.

For the dielectric model (Figure S1a), all A curves are nearly indistinguishable—, confirming that ε_r is dominated by the ~~Step~~ step 2 pore term (Eq. 3), not by A_{BN} . In porous NIPS films, ~~BN particle geometry is essentially irrelevant to~~ h-BN particle geometry has little influence on dielectric performance; only porosity level and pore architecture (A_{pore}) matter. This justifies treating the fitted dielectric A_{BN} as weakly determined (Table 1) and confirms that the RMSE values in Table 2 reflect mean-field model limitations rather than parameter uncertainty.

For the thermal model (Figure S1b), sensitivity to A is higher above 30 wt% ~~BN-h-BN~~, which is precisely why A was fixed at its literature platelet value rather than fitted (Section 3.4). At the NP mean porosity (65.6%), all $A = 0.3\text{--}6.0$ curves remain below the experimental NP 70 wt% value (0.16–0.18 vs. $0.305 \text{ W m}^{-1} \text{ K}^{-1}$), reinforcing that the 70 wt% enhancement is driven by the proximity to maximum packing and incipient particle contact, not by any plausible choice of single-particle shape factor. In the practically important 30–50 wt% range, the A curves converge within $\pm 0.02 \text{ W m}^{-1} \text{ K}^{-1}$, and the mean-field LN model is adequate.

4.7. ~~Mechanical Properties~~ Property trade-offs and ~~Coefficient of Thermal Expansion~~ design guidance

~~Figure ?? presents tensile strength and CTE results. Hot-pressing consistently improves tensile strength by 19–95% (Table 4), consistent with pore collapse under compressive load. The non-monotonic NP tensile strength—local maximum at 30 wt% (25.4 MPa), minima at 10 wt% (14.9 MPa) and 50 wt% (12.5 MPa)—correlates inversely with the anomalous porosity at 10 wt% (91.4%) and high porosity at 50 wt% (58.1%), respectively. The LN + Gibson–Ashby model reproduces this trend qualitatively.~~

~~(a) Tensile strength vs. h-BN content for NP and HP films ($n = 1$ specimen per bar). Percentage values indicate the NP→HP change at each composition; the 70 wt% slot carries a baseline stub because those films were too brittle to test, not because the strength is zero. (b) CTE vs. h-BN content for HP films ($n = 1$ per point; TMA, tension mode, 100–200 °C): measured values against the Turner lower bound and the rule-of-mixtures (ROM) upper bound, both computed with $\alpha_{\text{PT}} = 41.92 \text{ ppm/K}$ (measured). PCB~~

substrate target zone (3–17 ppm/K) is shaded. The measured CTE lies between the two bounds because the discontinuous load-path geometry of the porous sponge gives only partial elastic constraint (Section 3.3).

For CTE (Figure ??b), the Turner model (using $\alpha_{PI} = 41.92$ ppm/K as measured from the neat PI-HP sample) correctly predicts the neat-film value but substantially underpredicts CTE at all BN loadings (–61 to –80%). The ROM overestimates because it neglects elastic constraint entirely. The experimental CTE values (24.96 $\rightarrow$ 10.11 ppm/K from 10 $\rightarrow$ 70 wt%) lie between the two model bounds, physically consistent with partial elastic constraint in a porous sponge where h-BN platelets are embedded within discontinuous PI ligaments rather than a macroscopically continuous dense matrix. This porous-architecture effect is an important limitation of applying dense-composite CTE models to sponge materials and represents an opportunity for future model development incorporating open-cell foam mechanics. Critically, the measured CTE = 10.1 ppm/K at 70 wt% h-BN (HP) lies within the PCB-substrate compatibility window (3–17 ppm/K), confirming viability for copper-metallized flexible packaging applications.

Tensile strength and CTE data for PI/h-BN composite films: h-BN NP strength HP strength NP $\rightarrow$ HP CTE Turner ROM (wt%) (MPa) (MPa) (%) (ppm/K) (ppm/K) (ppm/K) 0 20.85 37.02 +77.6 41.92 41.92 10 14.86 17.74 +19.4 24.96 9.69 39.31 30 25.43 38.15 +50.0 16.49 4.00 33.43 50 12.47 24.28 +94.7 13.06 2.61 26.45 70 —*—*—10.11 1.98 18.02

*Not tested (film brittleness at 70 wt% h-BN). Turner and ROM use $\alpha_{PI} = 41.92$ ppm/K (neat porous PI, HP), $E_{PI} = 3.5$ GPa, $E_{BN} = 200$ GPa, $\alpha_{BN} = 1.5$ ppm/K, solid-phase

4.8. Property Trade-offs and Composite Design Guidance

For each h-BN loading, the calibrated LN model traces a continuous locus in ε_r – λ space as porosity is swept (Figure ??b4c); the 70 wt% HP state already lies within the target zone ($\varepsilon_r = 1.62$, $\lambda = 0.28$ W m^{–1} K^{–1}).

Three design insights emerge. First, along any fixed-composition locus, reducing porosity raises both ε_r and λ ; the two properties are fundamentally coupled through the same structural variable and cannot be independently tuned by porosity control alone; every porosity choice is a position on a trade-off curve. Second, increasing h-BN loading shifts the entire locus toward higher λ at any given ε_r , making composition the primary lever for thermal performance at constant dielectric specification. Third, the 70 wt% locus passes through the most favorable region: experimentally, the NP film at 54.2% open porosity ($\varepsilon_r = 1.28$, $\lambda = 0.31$ W m^{–1} K^{–1}) meets all three target levels defined below, and the HP film at 48.8% porosity ($\varepsilon_r = 1.62$, $\lambda = 0.28$ W m^{–1} K^{–1}) bracket an operating window in which all three target levels defined below are met or exceeded.

4.8. Model-Guided Computational Design Model-guided design: Performance Limits and Processing Prescriptions

A key advantage of a calibrated physics-based model over purely empirical data is the ability to perform systematic computational optimization — exploring The calibrated model can be interrogated across the full (h-BN loading, porosity) design space to identify theoretical performance limits and the processing conditions required to reach any specification. This constitutes model-guided composite design, rather than

requiring exhaustive trial-and-error experimentation, the calibrated 3-phase LN model provides a predictive surface that can be interrogated algorithmically across thousands of composition–processing combinations.

Figure ?? presents a three-part analysis. Panel (a) identifies the given specification. Figure 4c,d presents this analysis; the absolute levels of the model surfaces inherit the dielectric uncertainty quantified in Section 4.2.

The feasible processing window—defined as the porosity range simultaneously satisfying that simultaneously satisfies each of three progressively demanding specifications: Basic ($\varepsilon_r < 2.0$, $\lambda > 0.05 \text{ W m}^{-1} \text{ K}^{-1}$), Advanced ($\varepsilon_r < 1.8$, $\lambda > 0.08 \text{ W m}^{-1} \text{ K}^{-1}$), and Future ($\varepsilon_r < 1.6$, $\lambda > 0.12 \text{ W m}^{-1} \text{ K}^{-1}$), is shown in Figure S4. The Basic window is open across the entire composition range (even neat porous PI qualifies at $\sim 34\text{--}44\%$ porosity, since $\lambda_{\text{PI}} = 0.12 \text{ W m}^{-1} \text{ K}^{-1}$ exceeds the Basic threshold at moderate porosity); the Advanced window opens at $\geq 40 \text{ wt\% BNh-BN}$; and the Future window opens only at $\geq 64 \text{ wt\% BNh-BN}$, spanning $58\text{--}76\%$ porosity at $70 \text{ wt\%}$. Experimentally, the $70 \text{ wt\% NP}$ film ($\varepsilon_r = 1.28$, $\lambda = 0.31 \text{ W m}^{-1} \text{ K}^{-1}$) already satisfies the Future specification outright, and the $70 \text{ wt\% HP}$ film satisfies the Advanced specification; the model is conservative for ε_r at this composition (predicting 1.69 vs. the measured 1.28), so the experimental films outperform the model envelope—and the prescribed windows should therefore be read as sufficient, not necessary, conditions. Panel (b) Figure 4c shows the ε_r – λ trade-off loci for all loadings together with the global Pareto frontier of the material system, Pareto frontier computed over the full (loading, porosity) grid: the frontier coincides with the $70 \text{ wt\%}$ locus over essentially the entire accessible range, indicating the accessible range. Because the dielectric residuals exceed the spread of the data (Section 4.2) and the measured $70 \text{ wt\%}$ films depart from the predicted locus ($\varepsilon_r = 1.28$ measured vs. 1.69 predicted, NP), we read this as model-level consistency with the experimental finding that $70 \text{ wt\% h-BN}$ is the Pareto-optimal composition within the calibrated model. [editorial comment removed] Stars mark the recommended operating point on each locus (maximum subject to $\varepsilon_r \leq 1.8$). Panel (c) is the most favorable composition tested, not as a validated frontier of the material system beyond the measured range. Figure 4d answers the engineering question ‘what is the maximum λ achievable at a given ε_r ceiling?’: at $70 \text{ wt\%}$, the model envelope reaches $\lambda = 0.23 \text{ W m}^{-1} \text{ K}^{-1}$ at $\varepsilon_r \leq 1.6$ (porosity 58%), $0.31 \text{ W m}^{-1} \text{ K}^{-1}$ at $\varepsilon_r \leq 1.8$ (porosity 50%), and $0.39 \text{ W m}^{-1} \text{ K}^{-1}$ at $\varepsilon_r \leq 2.0$ (porosity 42%). The experimental $70 \text{ wt\%}$ points (NP: 0.305 ; HP: $0.279 \text{ W m}^{-1} \text{ K}^{-1}$) lie on or slightly above the $\varepsilon_r \leq 1.8$ envelope, validating consistent with the optimization surface in the experimentally probed region.

Model-guided design analysis. (a) Feasible processing window: the porosity range meeting each target specification vs. h-BN content (nested shading, light to dark for Basic→Future); measured NP and HP states overlaid. (b) trade-off loci for each h-BN loading; porosity swept along each curve; the $70 \text{ wt\%}$ locus (highlighted) lies above every other composition at every dielectric ceiling. The measured $70 \text{ wt\%}$ films sit above their own model locus, so the model is conservative here. (c) Maximum attainable under a given ceiling; porosity optimised at each point; stars mark the $\varepsilon_r \leq 1.6$ and $\varepsilon_r \leq 1.8$ operating points at $70 \text{ wt\%}$. (d) The porosity that maximises under each ceiling: the five composition curves stay within 10% (absolute

porosity) of one another, showing that the porosity target is set almost entirely by the dielectric ceiling while the achievable at that porosity is set by the loading. Panels (b)–(d) are model surfaces conditioned on the calibration of Table 1; see Section 4.2 for the residual analysis that bounds their reliability.

The optimization surface condenses into simple engineering prescriptions. The porosity required to maximize that maximizes λ under a fixed ε_r ceiling is remarkably flat — nearly independent of composition (42–50% for $\varepsilon_r \leq 1.8$ and 52–58% for $\varepsilon_r \leq 1.6$ across the entire composition range — meaning, Figure S5), so the porosity target is set almost entirely by the dielectric ceiling, while the achievable λ at that porosity is set by the BN-h-BN loading. In terms of specific specifications, achieving $\varepsilon_r < 1.7$ with $\lambda > 100 \text{ mW/m/K}$ $\lambda > 100 \text{ mW m}^{-1} \text{ K}^{-1}$ requires ≥ 60 wt% BN-h-BN (porosity 53–54%); $\varepsilon_r < 1.6$ with $\lambda > 120 \text{ mW/m/K}$ $\lambda > 120 \text{ mW m}^{-1} \text{ K}^{-1}$ requires 70 wt% at 58% porosity; and the most demanding specification ($\varepsilon_r < 1.5$, $\lambda > 150 \text{ mW/m/K}$ $\lambda > 150 \text{ mW m}^{-1} \text{ K}^{-1}$) is feasible only at 70 wt% and 63% porosity. Since the as-fabricated 70 wt% NP film already sits at 54.2% porosity, these prescriptions are reachable by mild or partial hot-pressing — full densification is neither required nor desirable, as since it would raise ε_r beyond the ceilings (and, per and, as shown in Section 4.3, risks disrupting the percolative BN network) h-BN contact network.

5. Conclusions

Porous PI/h-BN composite films were fabricated by NIPS and characterized in both over 0–70 wt% h-BN, characterized in the non-pressed (NP) and hot-pressed (HP) states were analyzed using a calibrated 3-phase states, and described with a calibrated three-phase Lewis–Nielsen framework. The following principal conclusions are drawn: NIPS-generated porosity is the dominant determinant of the dielectric constant: all films achieve lie at $\varepsilon_r = 1.28$ – 2.18 at 100 kHz (100 kHz), far below dense PI (~ 3.5), with and hot-pressing raising raises ε_r by 0.18–0.42 through pore collapse. The fitted pore shape factor $A_{\text{pore}} = 0.08$ quantifies the geometric inefficiency of NIPS bicontinuous sponge pores compared low field-shielding efficiency of the bicontinuous NIPS sponge relative to isolated spherical pores ($A = 1.5$), providing and provides a model-based microstructure-fingerprint that distinguishes NIPS from Pickering-emulsion-templated Pickering-emulsion-templated porous PI/BN composites. A deliberately parsimonious h-BN films. The dielectric residuals exceed the spread of the data, so the model serves as a mechanistic and design tool rather than as a quantitative predictor of ε_r for individual films.

A single-parameter thermal calibration ($\lambda_{\text{BN,eff}} = 6.81 \text{ W m}^{-1} \text{ K}^{-1}$, with platelet shape factor and packing fraction fixed at literature values; 7:1 data-to-parameter ratio) reproduces the thermal data with $\text{RMSE} \leq 0.03 \text{ W m}^{-1} \text{ K}^{-1}$. The $\approx 7\times$ conductivity data with $\text{RMSE} \leq 0.03 \text{ W m}^{-1} \text{ K}^{-1}$. The reduction of $\lambda_{\text{BN,eff}}$ well below bulk h-BN quantifies Kapitza interface the interfacial (Kapitza) resistance at 70 nm particle size, and the sharp increase in λ upturn at 70 wt% emerges naturally from the approach to maximum platelet packing ($\phi_{\text{BN}} = 0.59 \approx 0.99 \phi_{\text{max}}$) without an additional percolation parameter. Hot-pressing at 70 wt% reduces λ (0.305 $\rightarrow$ 0.279 $\text{ W m}^{-1} \text{ K}^{-1}$) despite reducing porosity — from 0.305 to 0.279 $\text{ W m}^{-1} \text{ K}^{-1}$.

opposite to the model trend—~~providing direct experimental evidence~~, indicating that compaction partially disrupts the ~~porelative BN-BN h-BN~~ contact network formed during NIPS. ~~Turner and ROM CTE models respectively underpredict and overpredict at ≥ 10 wt% BN~~ The measured CTE lies between the Turner and rule-of-mixtures bounds because the porous sponge breaks the continuous load-path assumption ;~~the experimental CTE lies between these bounds, reaching 10.1 ppm/K of dense-composite models, and reaches 10.1 ppm K⁻¹ at 70 wt% —(HP), within the PCB substrate compatibility window(3–17 ppm/K).~~ Global Pareto analysis shows that the window.

Within the compositions tested, 70 wt% h-BN locus constitutes the gives the most favorable ε_r – λ Pareto frontier of the entire material system. ~~The combination, and the calibrated model is consistent with this result: the 70 wt% NP film already satisfies the most demanding target examined ($\varepsilon_r < 1.6$, $\lambda > 0.12 \text{ W m}^{-1} \text{ K}^{-1}$) experimentally~~locus lies above every other composition across the accessible porosity range, and the model prescribes ~~the porosity window (a porosity window of 50–63% at 70 wt%, reachable by mild hot pressing from the as-fabricated 54%) required to navigate the trade-off toward any intermediate specification~~ by mild hot pressing. These results establish the pore shape factor as a microstructural design parameter for porous ceramic-filled polymer films and provide a calibrated basis for selecting composition and porosity in NIPS-derived PI/h-BN packaging films.

CRedit ~~Authorship Contribution Statement~~authorship contribution statement

~~[editorial comment removed]~~[Name 1]: Conceptualization, Methodology, Investigation, Formal analysis, Writing – original draft. ~~[editorial comment removed]~~[Name 2]: Resources, Supervision, Writing – review & editing, Funding acquisition. ~~Adjust roles as appropriate.~~[Adjust roles as appropriate.]

Declaration of ~~Competing Interest~~competing interest

The authors declare no competing financial interests.

Data ~~Availability~~availability

All experimental data underlying the figures are provided in the Supplementary Material and in the accompanying raw-data workbook. The complete Python implementation of the ~~3-phase~~ three-phase Lewis–Nielsen model, including the parameter-estimation protocol, is listed in ~~Supplementary~~ Section S5 of the Supplementary Material and archived in a public repository (~~[editorial comment removed]~~[DOI]).

Acknowledgments

Dielectric verification measurements were performed at the Korea Polymer Testing & Research Institute (KOPTRI). Thermal diffusivity measurements were provided by Korea University of Technology

and Education (KOREATECH). CTE measurements were carried out at Kyung Hee University. Mercury porosimetry and SEM/EDS were conducted at the Cooperative Equipment Center, [\[institution\]](#). The authors thank all collaborating institutions for access to instrumentation. ~~[editorial comment removed]~~
[\[Funding statement: grant number\(s\), or a statement that no external funding was received.\]](#)

References

- [1] R. Li, X. Yang, J. Li, Y. Shen, L. Zhang, R. Lu, C. Wang, X. Zheng, H. Chen, T. Zhang, Review on polymer composites with high thermal conductivity and low dielectric properties for electronic packaging, *Materials Today Physics* 22 (2022) 100594.
- [2] X. J. Liu, M. S. Zheng, G. Chen, Z. M. Dang, J. W. Zha, High-temperature polyimide dielectric materials for energy storage: theory, design, preparation and properties, *Energy & Environmental Science* 15 (2022) 56–81.
- [3] H. Li, F. Bao, X. Lan, S. Li, H. Zhu, Y. Li, M. Wang, C. Zhu, J. Xu, Fluorinated polyimide with triphenyl pyridine structure for 5g communications: low dielectric, highly hydrophobic, and highly transparent, *European Polymer Journal* 197 (2023) 112327.
- [4] Y. Guo, H. Qiu, K. Ruan, Y. Zhang, J. Gu, Hierarchically multifunctional polyimide composite films with strongly enhanced thermal conductivity, *Nano-Micro Letters* 14 (2022) 26.
- [5] W. Y. Tan, L. F. Jian, W. P. Chen, Y. W. Zhang, X. C. Lu, W. J. Huang, J. S. Zhang, J. W. Wu, A facile strategy for intrinsic low-dk and low-df polyimides enabled by spirobifluorene groups, *Chinese Journal of Polymer Science* 41 (2022) 288–296.
- [6] H. Li, F. Bao, S. Li, H. Zhu, Y. Li, M. Wang, C. Zhu, J. Xu, Preparation of fluorinated polyimides with low dielectric constants and low dielectric losses by combining ester groups and triphenyl pyridine structures, *Polymer Chemistry* 15 (2024) 22–29.
- [7] H. Li, X. Kong, S. Wang, M. Gong, X. Lin, L. Zhang, D. Wang, Sustainable dielectric films with ultralow permittivity from soluble fluorinated polyimide, *Molecules* 28 (2023) 3095.
- [8] S. Kim, J. Son, H. Park, E. Jeong, K. H. Nam, J. S. Bae, Polymer concentration and liquid–liquid demixing time correlation with porous structure of low dielectric polyimide in diffusion-driven phase separation, *Polymers* 14 (2022) 1425.
- [9] S. Khim, Y. C. Hwang, J. Choi, H. Park, K. H. Nam, Temperature-invariant large broadband polyimide dielectrics with multimodal porous networks, *ACS Applied Polymer Materials* 5 (2023) 4159–4169.

- [10] M. Kim, G. Kim, J. Kim, D. Lee, S. Lee, J. Kwon, H. Han, New continuous process developed for synthesizing sponge-type polyimide membrane and its pore size control method via non-solvent induced phase separation (nips), *Microporous and Mesoporous Materials* 242 (2017) 166–172.
- [11] W. Li, K. Li, W. Li, W. Gan, S. Song, Carbon nitride/polyimide porous film via an nips method with advanced dielectric and hydrophobicity properties, *RSC Advances* 14 (2024) 15270–15280.
- [12] W. Ren, X. Huang, G. M. Yan, J. Yang, G. Zhang, Structuring multiscale porous architecture in polyether sulfone films for ultra-low dielectric constant, *Polymer* 272 (2023) 125832.
- [13] X. Zhao, K. Ruan, H. Qiu, X. Zhong, J. Gu, Fatigue-resistant polyimide aerogels with hierarchical cellular structure for broadband frequency sound absorption and thermal insulation, *Advanced Composites and Hybrid Materials* 6 (2023) 171.
- [14] V. Guerra, C. Wan, T. McNally, Thermal conductivity of 2d nano-structured boron nitride (bn) and its composites with polymers, *Progress in Materials Science* 100 (2019) 170–186.
- [15] J. Gu, Z. Lv, Y. Wu, Y. Guo, L. Tian, H. Qiu, S. Zhu, C. Zhou, Dielectric thermally conductive boron nitride/polyimide composites with outstanding thermal stabilities via in-situ polymerization-electrospinning-hot press method, *Composites Part A: Applied Science and Manufacturing* 94 (2017) 209–216.
- [16] X. H. Ou, S. S. Chen, X. M. Lu, Q. H. Lu, Enhancement of thermal conductivity and dimensional stability of polyimide/boron nitride films through mechanochemistry, *Composites Communications* 23 (2021) 100549.
- [17] Y. Guo, Z. Lyu, X. Yang, Y. Lu, K. Ruan, Y. Wu, J. Kong, J. Gu, Enhanced thermal conductivities and decreased thermal resistances of functionalized boron nitride/polyimide composites, *Composites Part B: Engineering* 164 (2019) 732–739.
- [18] S. W. Jin, Y. J. Jin, Y. J. Choi, D. B. Kim, K. H. Yoon, H. W. Kim, C. M. Chung, Eco-friendly preparation and characterization of highly thermally conductive polyimide/boron nitride composites, *Composites Part A: Applied Science and Manufacturing* 166 (2023) 107396.
- [19] Y. Zhang, K. Ruan, K. Zhou, J. Gu, Controlled distributed Ti₃C₂Tx hollow microspheres on thermally conductive polyimide composite films for excellent electromagnetic interference shielding, *Advanced Materials* 35 (2023) 2211642.
- [20] K. Zhao, S. Wei, M. Wang, Y. Chen, Alkylated modified boron nitride nanosheets/polyimide composite films with advanced thermal conductivity and low dielectric constant, *Ceramics International* 49 (2023) 32577–32587.

- [21] K. Yang, Y. Y. Kang, H. J. Ahn, D. G. Kim, N. K. Park, S. Q. Choi, J. C. Won, Y. H. Kim, Porous boron nitride/polyimide composite films with high thermal diffusivity and low dielectric properties via high internal phase pickering emulsion method, *Journal of Industrial and Engineering Chemistry* 79 (2019) 154–163.
- [22] Y. Ma, Z. He, Z. Liao, Y. Han, J. Zhang, M. Zhu, Porous structure contained polyimide film with enhanced dielectric properties upon high temperature, *Journal of Applied Polymer Science* 139 (2022) e52936.
- [23] T. B. Lewis, L. E. Nielsen, Dynamic mechanical properties of particulate-filled composites, *Journal of Applied Polymer Science* 14 (1970) 1449–1471.
- [24] L. E. Nielsen, Thermal conductivity of particulate-filled polymers, *Journal of Applied Polymer Science* 17 (1973) 3819–3820.
- [25] R. C. Progelhof, J. L. Throne, R. R. Ruetsch, Methods for predicting the thermal conductivity of composite systems, *Polymer Engineering & Science* 16 (1976) 615–625.
- [26] P. S. Turner, Thermal-expansion stresses in reinforced plastics, *Journal of Research of the National Bureau of Standards* 37 (1946) 239–250.
- [27] L. J. Gibson, M. F. Ashby, *Cellular Solids: Structure and Properties*, 2nd Edition, Cambridge University Press, Cambridge, 1997.
- [28] J. Kim, G. Kim, S.-Y. Kim, S. Lee, Y. Kim, J. Lee, J. Kim, Y. C. Jung, J. Kwon, H. Han, Fabrication of highly flexible electromagnetic interference shielding polyimide carbon black composite using hot-pressing method, *Composites Part B: Engineering* 221 (2021) 109010. doi:10.1016/j.compositesb.2021.109010.
- [29] R. Kochetov, A. V. Korobko, T. Andritsch, P. H. F. Morshuis, S. J. Picken, J. J. Smit, Three-phase Lewis–Nielsen model for the thermal conductivity of polymer nanocomposites, in: 2011 Annual Report Conference on Electrical Insulation and Dielectric Phenomena (CEIDP), IEEE, 2011, pp. 338–341. doi:10.1109/CEIDP.2011.6232665.
- [30] J. Gu, K. Ruan, Breaking through bottlenecks for thermally conductive polymer composites: a perspective for intrinsic thermal conductivity, interfacial thermal resistance and theoretics, *Nano-Micro Letters* 13 (2021) 110.